MATHS AA HL · Regression and conditional probability

Question booklet · 35 minutes · 28 marks

Answer all questions. A calculator is permitted. Show your reasoning. Give exact answers or round to three significant figures unless directed otherwise. Write on separate paper.

1. Can a regression be reversed? [19]

Six ridge monitoring sites record exposure x and weathering index y. The paired observations are (2,3), (4,4), (5,7), (7,6), (8,9), (10,10). Use a GDC. A site is selected uniformly from these six.

(a) Find the means and the regression of y on x. Use it to predict y when $x = 6.5$. **[4]**

(b) Estimate x for $y = 8$ using the appropriate regression. Compare with inverting the first line. **[4]**

(c) Show that the product of the two fitted slopes agrees with the square of the correlation coefficient. Explain when the two regression lines would coincide for data with non-zero variances. **[4]**

(d) Let H denote an observed index at least 7 and E denote exposure at least 7. Calculate $P(H|E)$ and assess independence in this empirical distribution. **[4]**

(e) A colleague substitutes the fitted values for measured y before estimating $P(H|E)$. Find their answer and explain the discrepancy. **[3]**

2. What happens when the sample is reweighted? [9]

A survey has two regions. In region A, 60 of 100 shops sell an item and, among those 60, 24 offer delivery. In B, 20 of 50 sell it and, among those 20, 16 offer delivery. A sampling scheme first chooses A with probability w and B otherwise, then chooses one shop uniformly from the selected region. Let S mean the shop sells the item and D mean it sells the item with delivery.

(a) Derive expressions for $P(S)$ and $P(D)$ and hence for $P(D|S)$. **[3]**

(b) Find the choice of w for which half the selling shops offer delivery under this scheme. Compare with choosing uniformly from all 150 shops. **[3]**

(c) For $0 < w < 1$ determine whether region and delivery are independent given that the shop sells the item. Justify. **[3]**

End of question booklet · 28 marks